

BUSINESS INTELLIGENCE AND TOOLS

(24CSH-341)

EXPERIMENT 1: Data Acquisition and Profiling in Tableau

AIM

The goal of this experiment is to learn how to pull data from multiple different sources and use Tableau Desktop to profile it. By doing this, we want to check the data's quality, understand how it's structured, spot any weird anomalies, and get everything prepped for deeper analysis.

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OBJECTIVES

- Get a solid grasp of what Business Intelligence (BI) is and how data acquisition works.
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 - Learn to connect and pull together data from mixed sources, specifically CSV files, Excel spreadsheets, and SQL Server databases.
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 - Use Tableau's built-in tools to review the data's health and make sure it's ready to use.
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TOOLS & SOFTWARE REQUIRED

- Tableau Desktop (Tableau Prep Builder is a great optional add-on)

- Microsoft SQL Server & SQL Server Management Studio (SSMS)
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 - **Sample Datasets:** [Sales_Data.csv](#), [Employee_DB.xlsx](#), and a sample [Transactions](#) SQL database
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THEORY

What is Business Intelligence (BI)? At its core, Business Intelligence is all about the tools and processes we use to gather, clean, and analyze business data so we can make better, smarter decisions. A typical modern BI workflow involves a few key pieces:

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- **Data Sources:** Where the data lives (like databases, simple flat files, or cloud storage).
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- **ETL Pipeline:** The process of Extracting, Transforming, and Loading the data to get it ready for use.
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- **Analytics & Visualization:** Using tools like Tableau to turn that raw data into clear, actionable charts and dashboards.
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Bringing Data into Tableau Data acquisition is just the fancy term for gathering your data into one place. Tableau makes this easy through its **Connect** pane. Once you connect to your data, Tableau gives you two main choices:

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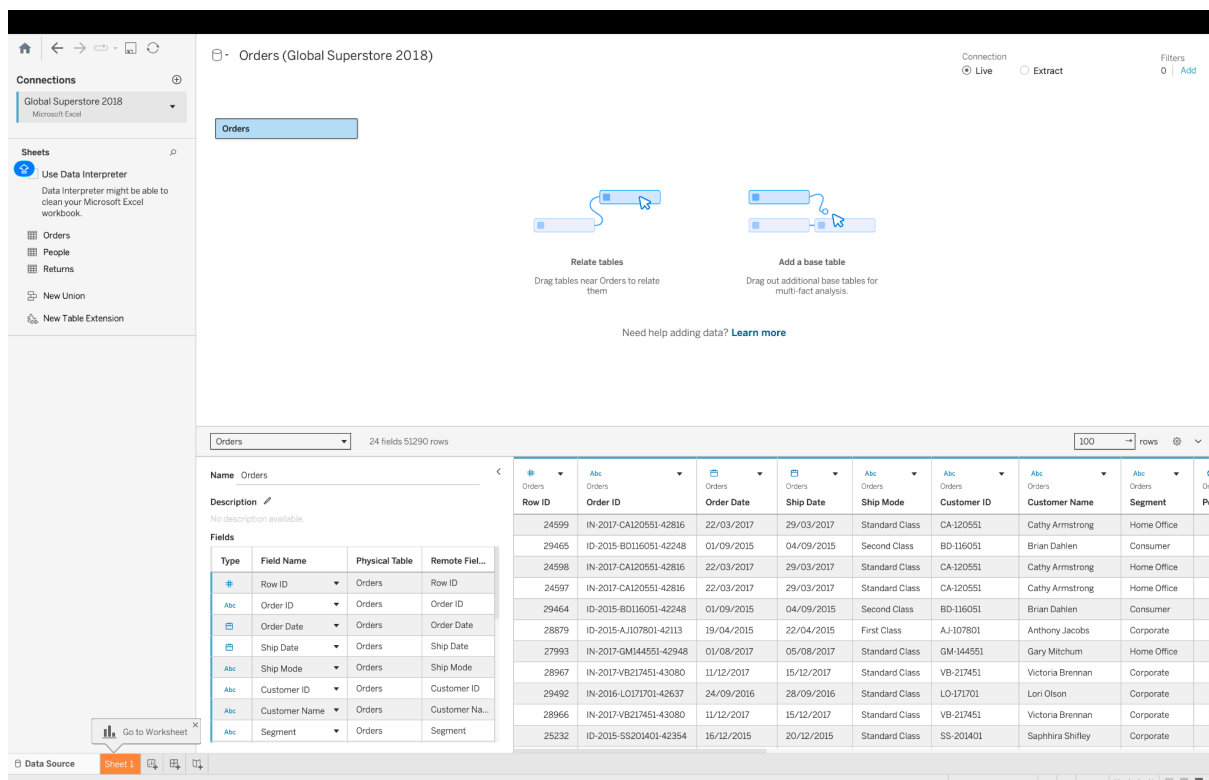
1. **Live Connection:** This queries your database or file in real-time.
2. **Extract Connection:** This takes a highly compressed snapshot of your data (saved as a [.hyper](#) file) and saves it locally, which usually makes your dashboards run much faster.

Profiling Your Data Before you start building charts, you need to know if your data is actually good. Data profiling is simply examining your dataset to check its completeness, accuracy, and overall health. In Tableau, you do this on the Data Source page by checking out the metadata grid (to make sure numbers are recognized as numbers, dates as dates, etc.) and looking for any obvious blanks or errors.

PROCEDURE

Part A: Setting Up and Connecting Your Data

- **Step 1: Get the Environment Ready**
 - Make sure Tableau Desktop is installed.
 - Open up SQL Server Management Studio (SSMS) and make sure your transaction database is running and accessible.
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- **Step 2: Bring in the CSV File**
 - Open Tableau Desktop. On the left side under the **Connect** menu, look for "To a File" and click **Text file**.
 - Find and open your **Sales_Data.csv** file. You'll see the data pop up on your screen.
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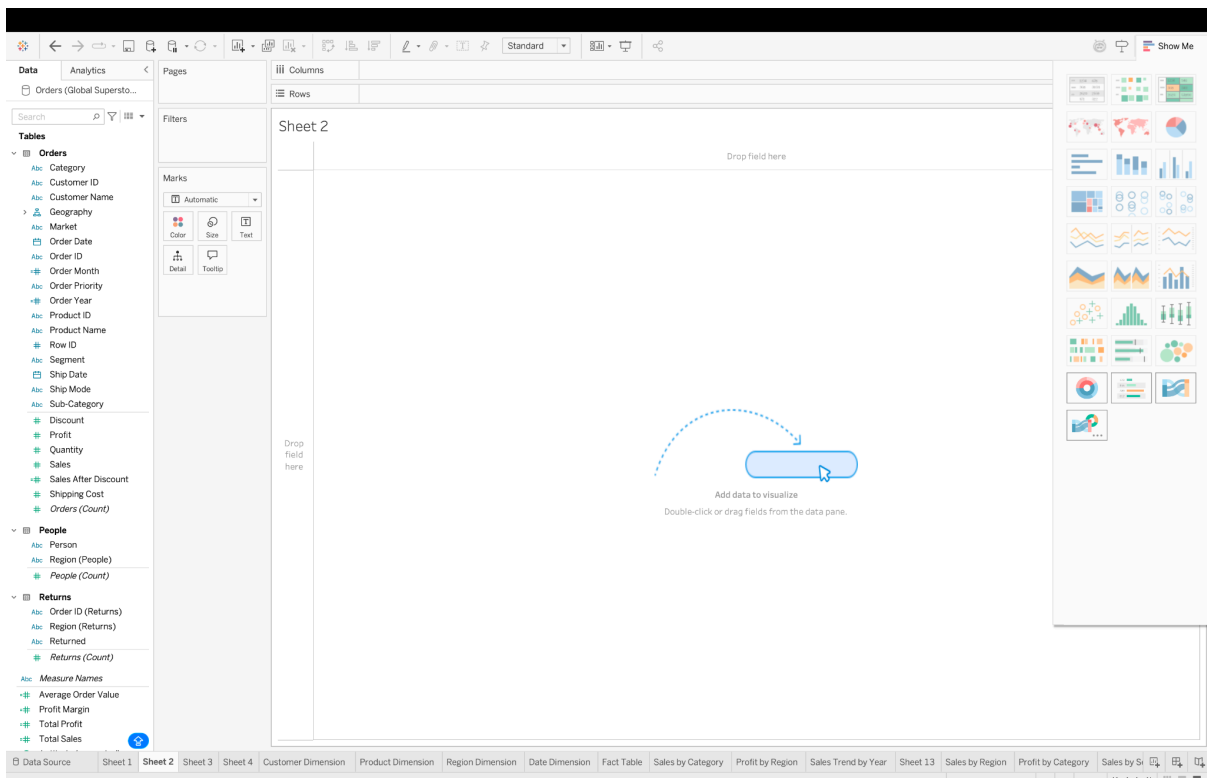
- **Step 3: Bring in the Excel File**
 - While still on the Data Source page, click the **Add** button next to "Connections" to bring in your next file.
 - Choose **Microsoft Excel**, find **Employee_DB.xlsx**, and drag the right sheet onto the main canvas.
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- **Step 4: Connect the SQL Database**
 - Click the **Add** button one more time, but this time look under "To a Server" and pick **Microsoft SQL Server**.
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- Type in your server details, connect to the database, and drag the **Transactions** table onto the canvas. Tell Tableau how these files relate to each other (for example, by linking them via a shared Employee ID or Transaction ID).
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Part B: Profiling and Checking Data Health

- **Step 5: Review the Data Source Page**

- Switch your connection type to **Extract** in the top right corner so Tableau can process the data quickly.
- Look at the data grid at the bottom of the screen. Toggle to the **Metadata Grid** view (the icon that looks like a list) to easily scan all your column names and data types.
- Check Tableau's icons above each column. Did it correctly guess that your Revenue column is a number (#) and your Zip Code is a geographic location (a little globe)? If not, click the icon to fix it.



- **Step 6: Spot the Issues**

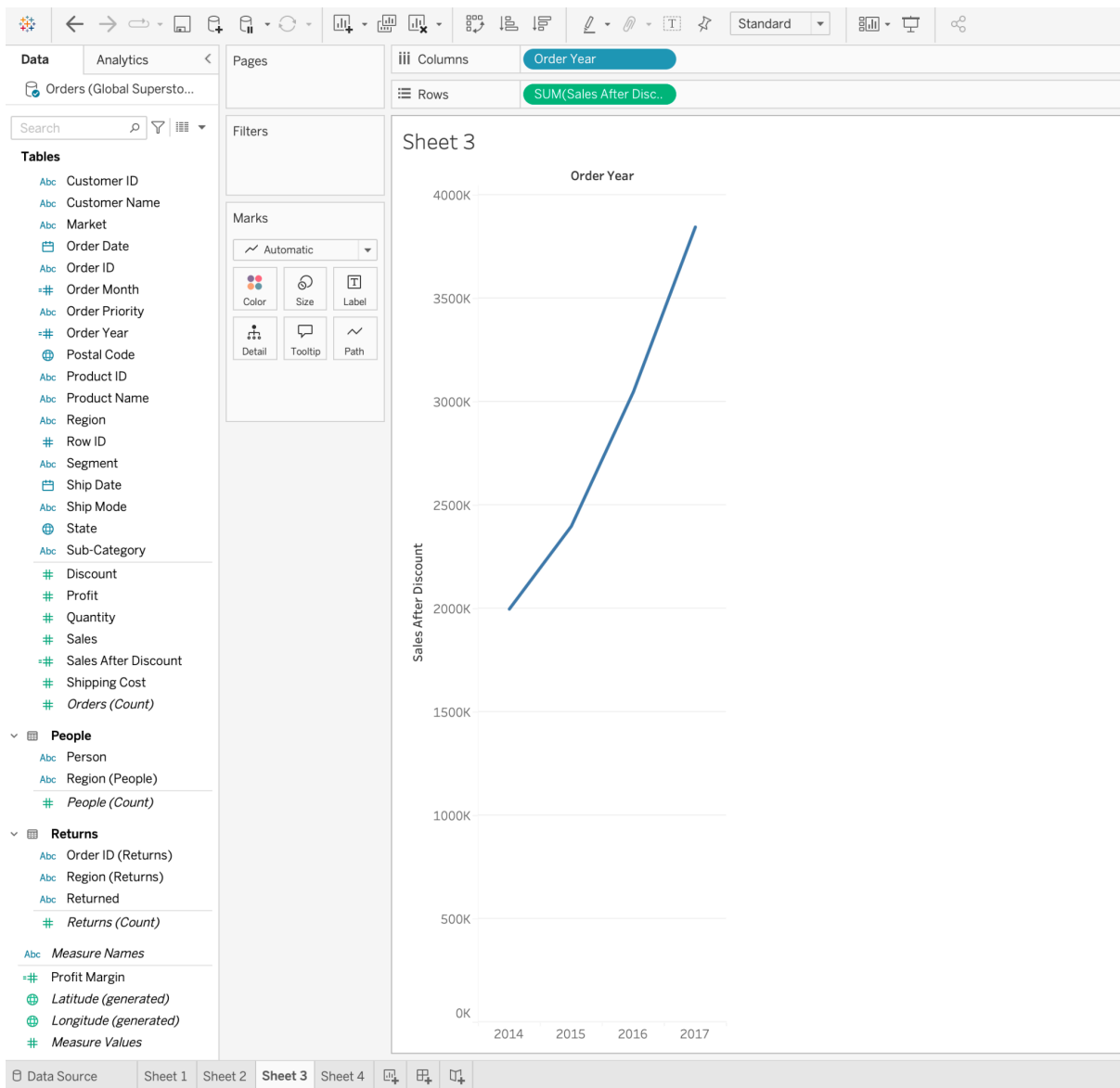
- Scroll through your data and look for red flags:
 - Are there a bunch of `Null` (empty) values in columns that shouldn't be empty?
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 - Are the date formats messed up?
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- Are there weird outliers, like a negative number for a sales price?

Part C: Documenting the Fixes

● Step 7: Record and Plan

- Write down at least three specific data quality issues you found during your review.
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- Jot down how you plan to fix them in Tableau—whether that means changing a data type right there in the grid, filtering out bad rows, or writing a quick formula to handle blank spaces.
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RESULT

We successfully connected Tableau to three completely different types of data sources: a CSV file, an Excel workbook, and a SQL Server database. Using Tableau's Data Source and Metadata grids, we profiled the data to check for missing values, validate formatting, and fix data type mismatches. This ensured our data was clean, integrated, and fully ready to be turned into visual dashboards.

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LEARNING OUTCOMES

- Understood how Business Intelligence works and why clean data is so important for making decisions.
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- Learned how to comfortably navigate Tableau Desktop's connection interface.
- Successfully brought together mixed data formats (CSV, Excel, SQL) into one unified data model.
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- Figured out how to spot data quality issues—like missing values or wrong data types—and correct them using Tableau's built-in tools.
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